

Cross-Border Private Equity and Valuation Effects*

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August 22, 2020

Preliminary. Please do not cite.

Abstract

This paper investigates the role of private equity (PE) firm valuations in cross-border leveraged buyouts (LBOs). Prior work shows that buyout leverage is increasing in PE fund performance, due to lower asymmetric information costs. Since fund Net Asset Value (NAV), as proxy for performance and a General Partner's skill, are affected by both portfolio-company related and economy-wide factors, I conjecture that cross-border LBOs are affected more by PE firm valuation changes relative to domestic LBOs. Extracting a quarterly measure of valuation from a fund's NAV, I document a strong and positive relationship between pre-acquisition valuation effects and probability of a cross-border buyout. Considering high levels of debt that characterize LBOs, the data reveals that favorable international financial conditions that signal 'Risk-on' states of the world, strengthen this positive association. Finally I find that the baseline results are particularly strong for low reputation PE firms for whom the asymmetric information costs in raising new capital are higher relative to high reputation PE firms. These findings are novel in the private equity literature.

Keywords: Private Equity, Leveraged Buyouts, Valuation

JEL Codes: G23, G32, G34.

*I would like to thank my advisor, Professor Anusha Chari for her invaluable insights, suggestions and encouragement. We thank workshop participants at the Macroeconomics Seminar at UNC-Chapel Hill for helpful comments and suggestions. I thank Greg Brown, Lutz Hendricks, Patrick Conway, Niklas Huther, Ayako Yasuda and Reiner Braun for helpful discussions.

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1 Introduction

Over the past four decades, private equity (PE) has grown into a sizeable asset class, with more than 10,000 funds raising in excess of \$ 2 trillion from institutional and other investors (Source: Preqin). Buyout funds account for 63 percent of this amount. Another notable trend is the rising share of cross-border buyout investments; figure 1 shows the number of cross-border buyouts as a share of total buyout transactions rose from 26 percent in 1991 to a high of 43 percent in 2017. In contrast to venture funds (which have received more academic attention) buyout funds usually purchase a controlling interest in an established corporation or one of its product lines, typically involving large amounts of debt (i.e., leveraged buyouts). Despite their important role in financing firms and reallocating capital to more productive sectors of the economy, relatively little is known about the investment behavior of buyout funds (Ljungqvist, Richardson and Wolfenzon, 2017).

In this paper, I examine the empirical relationship between one form of private equity investment, namely, leveraged buyouts (LBOs) and changes in PE firm valuations, or *Valuation Effects*. Prior work shows that private equity groups reduce the agency cost of lending through their general reputation (Demiroglu and James, 2010; Ivashina and Kovner, 2010; Achleitner et al., 2012) and performance (Demiroglu and James, 2010; Malenko and Malenko, 2015), leading to more favorable borrowing terms. It follows then at the time of lending to a potential target, banks must rely on observable PE firm characteristics such as the Internal Rate of Return (IRR) and Net Asset Value (NAV) as signals of the sponsor's past performance and reputation¹. Since the IRR and NAV are subject to both fund-specific and economy-wide shocks, I suggest changes in valuations impact the likelihood of a cross-border LBO more, relative to a domestic LBO. Following Froot and Stein (1991) and Erel, Liao and Weisbach (2012), I conjecture that when a foreign PE firm's valuation increases relative to that of a domestic one due to country-level macroeconomic differences, it can bid more aggressively for the target than domestic rivals because of a reduction in the magnitude of

¹The NAV a private equity fund is defined as the sum of the NAVs of its unrealized portfolio companies

information problems it faces in raising new capital.

This paper considers three questions: Do PE firm valuations affect cross-border buyout probabilities more relative to domestic buyouts? Do global financial conditions modulate this relationship between PE firm valuations and cross-border buyouts? Considering low reputation PE firms face higher asymmetric information costs, are valuation effects stronger for low reputation PE firms? To address these questions, I use a global dataset on 11,000 private equity buyout transactions consisting of 654 PE firms from 1994 to 2018, fund-level cashflow and NAV and target firm financial variables.

To set the stage, I begin by verifying the importance of valuation effects using quarterly change in IRR on the probability of a cross-border buyout relative to that of a domestic buyout. The data reveals an increase in the probability of a cross-border buyout in the quarter following a positive change in IRR. Since the IRR is heavily dependant on the timing of cash flows into or out of the fund (Jenkinson et al., 2013), I extract a measure of valuation directly from a fund's NAV. Following Jenkinson et al. (2013), Brown, Gredil and Kaplan (2016) and Barber and Yasuda (2017) I define valuation effects as the quarterly change in fund NAV adjusted for cashflows. Since lenders will assess firm-level performance (as proxy for skill and reputation) I aggregate this valuation measure to the firm-quarter level using fund-size as weights.

Next, I estimate a logit model of cross-border buyout probabilities as a function of pre-acquisition valuation effects while controlling for variation in PE firm, target firm and external environment characteristics. Consistent with the idea proposed, I find a strong and positive relationship between PE firm valuation and probability of a cross-border buyout relative to a domestic buyout. Several iterations robust to alternate deal and target-firm variables as well as acquirer and target country characteristics confirm my hypothesis. This finding is novel.

I also conjecture that shocks to global financial conditions modulate this relationship between valuation effects and cross-border buyout probabilities. Asis, Chari and Haas (2020)

suggest that a cost of external finance channel operates through different types of global funding shocks. Candidate global financial shocks that receive attention in the capital flows literature primarily falls into three inter-related categories: (i) U.S./advanced economy monetary policy shocks, (ii) shocks that impact foreign investor risk aversion and liquidity in international financial markets, and (iii) exchange rate shocks. Consistent with their idea, I conjecture that an easing of global funding conditions strengthen the positive association between rising valuations and cross-border buyout probability. The data reveals that a flatter US yield curve, higher excess stock market return over 1-month treasury rate, depreciating broad dollar, falling TED spread and lower VIX strengthen the impact of valuation changes on the probability of cross-border buyouts. To formally characterize how changes in risk attitude govern cross-border buyouts, I construct a "Risk-on", "Risk-off" index and show that the probability of cross-border buyout is higher following positive valuation effects in 'Risk-on' states of the world.

Finally, I examine how reputation modulates the impact of valuation effects on cross-border buyout probability. Consider, small young General Partners (GPs) that lack top performing funds in their track record. A positive NAV shock will be more important for these GPs, as they have little track record or accumulated reputation capital to obtain favorable loan conditions. To test this conjecture, I define low reputation PE firms in terms of performance, frequency of interaction with lending markets as well as size. High reputation PE firms are the complements to low reputation PE firms. I show that cross-border buyout probabilities are stronger for low reputation PE firms as positive valuation shocks reduce the agency cost of lending more for these types of firms relative to high reputation firms. Overall the results in this paper suggest that, due to country-level macroeconomic differences, valuation effects are particularly relevant for cross-border LBOs relative to domestic LBOs. These effects are also shaped by changing global financial conditions and heterogeneity in PE firm reputation.

The paper proceeds as follows. Section 2 surveys the related literature and discusses

the contribution to the private equity as well as the cross-border M&A literature. Section 3 provides a more detailed discussion of the economic transmission mechanism of valuation effects and cross-border buyout investments. Section 4 discusses the data used in the analysis. Section 5 reports the main results and associated discussion. Section 6 concludes.

2 Related Literature

This paper primarily contributes to the literature on investment behavior of private equity buyout funds. Malenko and Malenko (2015) show that buyout leverage increases with PE sponsor's skills. Their model of LBO activity proves that if PE firms never add any operational value to portfolio companies, no buyout takes place suggesting higher PE firm performance leads to more favorable borrowing conditions in subsequent deals. Ljungqvist, Richardson and Wolfenzon (2017) develop a model of buyout fund investment behavior based on imperfectly competitive market for private equity in which demand varies over time and supply is sticky in the short run. An important element of their model, consistent with the idea proposed in this paper, is that fund manager's can affect the market's perception of their talent by generating high returns in their current funds. They find that funds generally make acquisitions when investment opportunities are good, bargaining power is higher and debt is cheap.

My results also complement those in two highly influential papers in the cross-border M&A literature. In their canonical paper, Froot and Stein (1991) show that when a foreign firm's value increases relative to that of a domestic one, for example, through unhedged exchange rate changes or stock market fluctuations, its cost of capital declines relative to that of a domestic firm because of a reduction in the magnitude of information problems it faces in raising new capital. Because this explanation for a relation between valuations and cross-border acquisitions is based on asymmetric information, it is likely to be particularly relevant in the case of private targets for which asymmetric information tends to be high

relative to otherwise similar public targets (Erel, Liao and Weisbach, 2012).

Erel, Liao and Weisbach (2012) argue that what little work has been done on cross-border acquisitions focuses on public and/or US firms. However, they suggest most cross-border merger and acquisitions do not involve U.S. firms and do involve privately held firms. In their global sample of 56,978 cross-border acquisitions that occurred between 1990 and 2007, 97 percent involved a private firm as either acquirer or target while 53 percent did not involve U.S. firms. They suggest that even though cross-border acquisitions may be occurring for the same synergistic reasons as domestic acquisitions, country-level factors such as currency appreciation and macroeconomic performance as captured by stock market movements are likely to be making these acquisitions significantly more attractive for the acquiring firm. Consistent with prior literature, they find that countries with relatively superior-performing stock markets tend to purchase firms in the worse-performing country. Furthermore, countries whose currencies have appreciated are more likely have acquiring firms while countries whose currencies have depreciated are more likely to have target firms. They conclude that higher acquirer valuations lead to a higher propensity of cross-border acquisitions.

My work is distinguished from these papers in three main ways. First, to the best of my knowledge, the literature does not provide aggregate or micro evidence on the link between valuation effects and cross-border buyout investments, choosing to focus on strategic acquisitions (Erel, Liao and Weisbach, 2012) or aggregate FDI (Froot and Stein, 1991). This paper extracts a measure of valuation from private equity fund-level data to examine the relationship between valuations and cross-border buyout probabilities. The granularity of our data allows use to overcome shortcomings of country-level measures previously used in the M&A literature which ignore firm-level information. I conjecture that rising PE firm valuations, as proxy for reduced information asymmetry costs, raises probability of cross-border buyouts. I elaborate on this transmission mechanism in the next section.

Second, this paper extensively analyzes the importance of global financial conditions in

modulating the relationship between PE valuations and cross-border buyout probabilities. Axelson, Jenkinson, Stromberg and Weisbach (2007) document on average 75 percent of purchase price in leveraged buyouts are funded with debt. I draw on the capital flows literature to document how global financial shocks measured by U.S/advanced economy monetary conditions, shocks that impact foreign investor risk aversion and liquidity in financial markets and exchange rate shocks affect this link between cross-border buyout probability and PE firm valuations. Third, I uniquely emphasize heterogeneity in PE firm reputation in regulating the impact of valuation changes on cross-border buyout probability. I hypothesize that low reputation PE firms face problems of asymmetric information to a greater extent relative to high reputation firms. As a result, following a positive valuation shock the probability of a cross-border buyout is greater for these types of firms relative to high reputation firms.

3 Cross-Border Buyout and Valuation Effects: Transmission Mechanism

This section surveys the transmission mechanism through which valuation effects may impact the probabilities of a cross-border buyout. The mechanism relies on two key assumptions: (i) changes in fund valuations proxy changes in asymmetric information costs in raising debt, and (ii) Fund NAV and IRR are affected by both portfolio company performance (fund-specific or firm-specific effect) as well as macroeconomic changes (country-level effect).

The relationship between a PE firm and a bank is unique because the formal loan contract is between the portfolio company and the bank, with no direct claim against the PE firm. However, PE firms are effectively shadow borrowers as they control the borrower's equity, management, capital structure and strategic decisions (Ivashina and Kovner, 2011). Prior studies argue that, since targets are held privately by PE firms with limited public disclosure, PE firms mitigate asymmetric information costs in lending markets through their general reputation (Achleitner, Braun, Hinterramskogler and Tappeiner, 2012) and past performance

(Demiroglu and James, 2010; Malenko and Malenko, 2015).

To fix ideas consider the following. In a typical LBO, private equity sponsors present the potential deal to banks with a view to obtaining debt. Axelson, Jenkinson, Stromberg and Weisbach (2007) documented that on average 81.3 percent of non-equity financing in leveraged buyouts is bank debt ². The relationship between General Partners (GPs) in a private investment firm and outside lenders is complicated by lack of hard and timely information on a potential target company. Consequently, when making lending decisions, banks must rely on the reputation of sponsoring PE firms through observable performance indicators such as past IRR and NAVs which signals the GP's talent or skill in selecting, monitoring and restructuring target companies (Demiroglu and James, 2010) ³. Lending to PE firms with better performance and higher valuations could rationally be viewed as less risky by lowering asymmetric information costs, leading to better lending terms⁴.

Next consider the implications of assumption (ii). Froot and Stein (1991) and Erel, Liao and Weisbach (2012) suggest country-level macroeconomic differences are additional sources of valuation effects for cross-border acquisitions relative to domestic acquisitions. More formally, valuation effects arise from both unrealized portfolio holdings (fund-specific effects) or economy-wide changes (country-specific effects). Consider two PE firms of similar characteristics bidding for a potential target: one PE firm is domiciled in Country A, the other in Country B. Assume the target is also domiciled in country B. Suppose country A experiences some exogenous macroeconomic upswing resulting in upward valuations of all domestic portfolio holdings of PE firm in country A. Higher valuations, as a proxy for lower information asymmetry costs, raise access to bank debt allowing PE firm in country A to bid more aggressively for the target, raising the probability of a cross-border buyout.

²It typically is broken into tranches, with the average deal containing 3.8 tranches. Senior bank debt provides 52 percent of the debt financing, junior bank debt 4.7 percent, and 'contingent' debt such as a revolving credit facility 24.6 percent

³For example, private equity group involvement could lead to more rapid management turnover following poor operating performance. See Cimilluca (2007)

⁴Academic evidence suggests relying on past performance as a predictor of subsequent performance is reasonable. For example, Kaplan and Schoar (2005) document that absolute and relative performance of earlier funds predicts that of subsequent funds managed by the same private equity firm.

In sum, I conjecture that an increase in fund valuations, observable through fund-level NAV and IRR, will lead to higher probability of a cross-border deal relative to a domestic deal ⁵.

Closely related to the use of NAV as a measure of valuation is a growing literature on NAV manipulation by GPs. Various studies have focused on agency problems around fundraising, centering on the question of whether agents inflate portfolio values prior to fundraising (Eg: Jenkinson, Soussa and Stucke (2013); Barber and Yasuda 2017; Brown, Gredil and Kaplan (2016)). These studies use fund-level data and find that fund returns are systematically lower after fundraising events. Alternatively, Huther (2016) uses portfolio-company level data and finds no evidence of NAV manipulation and suggests that lower returns after fundraising is a result of deal composition. In other words, he argues it reflects true economic value of the underlying portfolio companies. For the purposes of this paper, I expect to observe cross-border buyouts following changes in valuation, regardless of whether they occur due to manipulation or true changes in economic value.

4 Data

4.1 Data Source

I use deal and fund-level data from Preqin. To obtain fund-level data, Preqin gathers information from public sources and make direct requests for submission. Freedom of Information Act (FOIA) requests (or their parallel outside the U.S.) requiring LPs to reveal certain information are often used, combined with requests to both LPs and GPs to voluntarily make their information public (Brown et al, 2015)⁶.

⁵Previous studies have also used fund size as a measure of reputation. Several papers show that PE fund IRR and NAV determine future capital commitments. See for example, Jenkinson, Soussa and Stucke (2013); Brown, Gredil and Kaplan (2016); Barber and Yasuda (2017). Since fund performance drives future capital commitments or fund size, valuation measures are reasonable approximations of reputation

⁶A downside of the public approach is that reliance on FOIA disclosures and voluntary submissions may lead to a sample that is not representative of the universe of funds. FOIA taps only certain types of investors (e.g., public pension funds); moreover, voluntary submission (especially by GPs) may introduce selection and

Preqin’s fund-level data is considered a representative sample in the literature. For instance, Brown et al (2015) examine fund performance across various datasets. They find that performance based on Preqin data is comparable to other widely used datasets. They also find that Preqin’s sample of geography, fund count and committed capital is also comparable to other datasets. Recent papers which have used Preqin data include Easton, Larocque and Stevens (2020), Ang, Chen, Goetzmann and Phalippou (2018), Barber and Yasuda (2017) and Ewen, Jones and Rhodes-Kropf (2013).

My analysis rests on three datasets from Preqin: (i) Deals dataset and (ii) Fund-level quarterly cashflows and NAV, (iii) Performance Analyst Database which provides the net private equity fund performance and performance benchmarks, as well as fund type, vintage year and size. The first advantage of using fund-level data is that it removes endogeneity concerns related to country-level valuation measures and allows researchers to extract a valuation effect directly from a fund’s Net Asset Value. A second advantage comes from the quarterly frequency of the dataset, providing a rich time-series dimension to exploit.

First, I retrieve fund-level quarterly data on IRR, Size and Investment Multiple from the Performance Analyst Database. The Investment Multiple is the ratio of all fund distributions and remaining NAV (i.e. "Total Value") to total capital calls (i.e. "Paid-in Capital"). Second, I retrieve deals data. Preqin’s deal-level data contains information on leveraged buyout, add-on, venture and growth capital investments. Each observation is a deal that contains name of the acquiring private equity firm, the name of the portfolio company, their respective countries of domicile, the deal value, the portfolio company’s pre-acquisition revenue, primary industry and Entry Multiple (Enterprise Value/Sales). I retrieve all buyout investments from 1990 to December 2019 and drop add-ons and growth capital investments even if they were made by a buyout fund since my focus is only on buyout deals, which typically utilize a large proportion of debt⁷. I keep only completed deals, as opposed to

survivorship biases. Moreover, this process may make it more difficult to check the data for accuracy or create comprehensive performance histories.

⁷Even though add-on investments also use debt, I drop this type of deal since my data does not provide platform company valuations.

announced or rejected bids and also drop any observation for which the year, acquiring firm country or portfolio company country is not available. After this screening process, I am left with a sample of 39,617 deals.

4.2 Constructed Variables and Controls

Next, I retrieve Private Equity Cash Flow Data which provides full cash flow information (calls, distributions, and quarterly NAVs) at the fund-level and is the key data that allow me to measure valuation effects for sample firms. All cash-flow information and NAVs are scaled by fund size and represent a hypothetical LP capital commitment of USD 10,000,000. As Brown et al (2016) and Jenkinson et al (2013) suggest, popular measures of performance such as the Internal Rate of Return (IRR) have significant drawbacks if used as a measure of valuation. The IRR is heavily dependant on the timing of cash flows into or out of the fund, and so will also be dependant on the holding period of an investment (Jenkinson et al, 2013).

Using Preqin’s quarterly cash flow and NAV data, I construct fund-level cash-flow adjusted valuation effects drawing on ideas suggested in three recent papers: Jenkinson et al (2013), Brown et al (2016) and Barber and Yasuda (2017). Jenkinson et al (2013) note that the NAV of a fund will change for one of three reasons: (i) cash is called from investors which is used to invest in a portfolio company, (ii) if cash is distributed back to investors when dividends are paid from existing portfolio companies or they are sold and (iii) the valuation of an existing portfolio company changes due to its performance and/or changes in the market valuation of companies.

I isolate this third factor as it captures fund-specific changes or economy-wide changes such as aggregate stock market movement or currency appreciation. Following Barber and Yasuda (2017), *Valuation Effects* are defined as NAV changes adjusted for calls (C_{jit}) and

distributions (D_{jit}) for each fund j in firm i in year-quarter t :

$$\Delta NAVCF_{jit} = NAV_{jit} - (NAV_{jit-1} + C_{jit} - D_{jit}) \quad (1)$$

Since I observe buyout deals at the PE firm-level, valuation effects are aggregated at the firm-quarter level using fund-size weighted average of all funds' valuation effects for a given firm-quarter⁸. To the extent that lenders assess the PE firm's performance and reputation at the firm-level (as opposed to the fund-level), aggregating to the firm-quarter level is not only reasonable but necessary for the theory underpinning my analysis.

Next, I match firm-level valuation measures (cashflow adjusted NAV change and IRR change) to deals data and augment this data with a set of controls which consists of three types of variables: PE-firm specific control, portfolio-company specific accounting variable and deal-specific controls. PE-firm specific control is what I define as *Experience* which is simply the firm's age; portfolio-company specific accounting variable is company revenue; deal-specific variables are: Enterprise Value/Sales and deal value. Although I would have liked to include a variable on portfolio company profitability (Eg: Earnings before Interest, Taxes, Depreciation and Amortization (EBITDA)) or a valuation multiple based on profitably (Eg: Enterprise Value/EBITDA), unfortunately good quality data are not available for the firms in my sample.

Finally, as suggested in Asis, Chari and Haas (2020), the globalization of and increased interconnectedness of financial markets propagates the transmission of financial and economic conditions between developed countries as well as from developed to emerging markets. Due to their reliance on debt markets, I suggest that buyout firms are likely to be affected by changes in global financial conditions. For this reason, I include several global variables that may affect the relationship between valuation effects and cross-border buyout probabilities. Similar to Asis et al (2020), my first proxy for international debt market conditions is the U.S.

⁸Brown et al (2016) construct change in Public Market Equivalent (PME) as a valuation measure using NAV changes adjusted for cashflows. They also adjust for market returns, which we do not do since our purpose is to capture the relationship of market returns and buyout investments at the firm-level

yield curve slope calculated as the difference between the 5-year treasury and the Fed funds rates, retrieved from Thomson Reuters. My second global macro variable is Excess Market Return, calculated as the difference between the value-weighted return on all NYSE, AMEX and NASDAQ stocks and the one-month treasury bill rate, retrieved from Ken French's website. The third global variable, the Chicago Board Options Exchange's CBOE Volatility Index (VIX), captures shifts in foreign investors' risk appetite in risk-on risk-off states of the world. Next, to capture liquidity and exchange rate shocks, I add TED Spread and Broad Dollar Index from the FRED, Federal Reserve Bank of St. Louis.

Finally, to measure changes in risk attitudes, I construct a Risk-on, Risk-off (RORO) index similar to that described in Datta et al (2017). The RORO index comprises of daily changes in several standardised variables, retrieved from FRED: ICE BofA BBB Corporate Index Option-Adjusted Spread, Moody's Seasoned Aaa Corporate Bond Yield Relative to Yield on 10-Year Treasury Constant Maturity, Moody's Seasoned Baa Corporate Bond Yield Relative to Yield on 10-Year Treasury Constant Maturity, Broad Dollar Index, TED spread, NASDAQ Index, S&P 500 Index, EURO STOXX Index, CBOE Volatility Index, NASDAQ Volatility Index, CBOE DJIA Volatility Index and Gold spot price. First, I normalize components such that positive changes imply risk-off behavior. Then, I scale these normalized changes by their respective historical standard deviations. After these transformations, the index is a simple average of the components⁹.

The number of PE-firm-quarter and deals with data for any variable is 39,617. However, due to missing data and sparsity of fund-level financial data, my final matched sample includes 11,252 buyout observations, with 3,801 cross-border buyout deals and 654 distinct PE firms.

⁹To match my quarterly valuations data, the index is aggregated to the year-quarter level using simple average

5 Methodology and Results

5.1 Univariate Statistics

Before examining my main research questions, I provide in Table 1 a descriptive analysis of private equity firm and portfolio company characteristics. Rows 1 and 2 of Panel A report fund IRR and fund size in the full Preqin sample. Mean IRR is 11.16, broadly consistent with Brown et al (2015) and higher than the median IRR of 8.83. Fund size values display large dispersion with a mean size of USD 992 million, 2.5 times greater than the median fund of USD 389 million. This can be attributed to the emergence of mega-funds: column 7 of row 2 shows funds at the 95th percentile are valued at USD 3.8 billion.

Focusing on the matched sample, row 3 shows firm-level capital commitments. The mean and median values echo patterns similar to those at the fund-level reported in row 2, with mean firm capital commitments higher than the 75th percentile. The next two rows shows weighted-average firm-level IRR and change in weighted-average firm-level IRR, where the weights are fund-size. The patterns in weighted-average firm-level IRR echo those at the fund-level for the full Preqin sample. For example, the mean firm-level IRR is 11.83, similar to the mean fund-level IRR of 11.16.

I next move onto examining NAV distribution. At any given point in the life of a fund, the NAV represents the sum of its cash contributions from limited partners to date (Capital Calls), less any distributions of realized returns to limited partners to date (Distributions), plus accounting recognition of estimated unrealized returns made on investments within the fund, net of fees. Thus, in a given quarter, *Valuation Effect* is the change in quarterly NAV that is not explained by contributions, distributions and fees during the quarter. Consistent with this idea, rows 6 and 7 show fund-size weighted firm-level NAV and change in NAV adjusted for calls and distributions. The mean fund-weighted Firm-level NAV is \$4,856,250 on a scaled LP capital commitment \$10,000,000¹⁰. Considering the differences in fund size,

¹⁰The ratio of NAV to capital commitments is approximately similar to that in Barber and Yasuda (2017)

I observe large dispersion in *Valuation Effects* with mean values significantly higher than the median. Turning to the Investment Multiple, I note that the mean fund-size weighted firm-level multiple is 1.30, approximately equal to the median multiple. Firm-level Investment Multiple at the 75th and 95th percentile is 1.52 and 2.35 respectively. Finally, row 9 reports private equity firm age, which captures experience, computed as the difference between the date of firm inception and a buyout deal completion. Firms age at the 25th and 75th percentiles are 13 and 28 respectively.

Looking at portfolio firm characteristics, Panel B of Table 1 reports pre-acquisition Revenue, Enterprise Value/Sales and Deal Value for the matched sample. Portfolio firm size displays large variation: the median firm reports revenue of USD 250 million, considerably less than the mean revenue of USD 5.45 billion. Even the 75th percentile in sample revenue is significantly less than the mean at USD 816 million. This should not be surprising considering the differences in fund-size observed in Panel A. Consistent with this pattern, median Enterprise Value/Sales is 1.48, considerably less than the mean of 3.97. Given the differences in portfolio firm size, deal values display similar patterns.

Finally, to get a sense of coverage across countries, Table A1 in the Appendix lists the top 15 acquirer countries in buyout investments from 1990 to December 2019. These 15 countries together account for 96 percent of total worldwide buyout deals and 90 percent of worldwide cross-border buyouts. Somewhat intriguingly, I find that the share of cross-border deals with US acquirers is lowest among the top 5 countries at 23.2 percent. Cross-border deals conducted by PE firms from UK account for 56 percent of total deals conducted by UK. I also observe that China is the only country outside of the developed world that enters the top 15 acquiring country list.

Table A2 in the Appendix lists the top 15 countries that receive the highest number of buyout investments from 1990 to May 2020. These 15 countries together receive 74 percent of worldwide buyout deals. I observe that the share of cross-border buyouts in total buyout

activity in the US is the lowest among the top 20 countries at 5.4 percent ¹¹. In other words, an overwhelming share of buyout activity in the US are conducted by domestic PE firms. Emerging market economies like India, China and Brazil also receive substantial number of buyout deals. China receives the largest number of buyout deals in the developing world. Their share of cross-border deals as targets is 88 percent, signifying growing global interest in the Chinese economy.

5.2 Cross-border buyouts and Valuation Effect

Next, I move on from inspecting summary statistics to examining cross-border buyout in detail to formally test for the statistical significance of the patterns suggested earlier. My goal is to test the hypothesis that rising acquirer valuations can lead to higher buyout investments since it relaxes agency costs of debt by reducing asymmetric information. Following Erel et al (2012), I assume a logistic distribution for the marginal probability of a buyout being cross-border, relative to being domestic, for a given quarter:

$$P_t(Y_{it}) = \frac{1}{1 + \exp(-\alpha - \beta_1 \frac{1}{n} \sum^n x_{ij,t-n} - \gamma z_{i,t})} \quad (2)$$

The dependant variable is assigned a value of 1 if PE firm i in country j in year-quarter t acquires a cross-border company, and 0 if it acquires a domestic company. My primary interest lies in the coefficient β_1 which measures the marginal change in the likelihood of a cross-border deal in year-quarter t due to a unit change in the acquirer's valuation effect in year-quarter $t - n$. I choose to use the average of the two quarters preceding acquisition for two reasons: (i) It creates an appropriate lag effect that would otherwise be absent for deals that were completed at the beginning of any year-quarter since valuation changes are based of end-of-quarter values (ii) Announcements can be made at least one quarter prior to

¹¹I found this number lower than my priori, hence I verified that my original unmatched buyout deals data from Preqin shows a similar pattern. I found that the share of cross-border deals targeting US firms is 7.1 in my unmatched full Preqin dataset, quite comparable to the matched sample.

deal completion, hence I reduce the announcement effect on valuation by averaging valuation effects in year-quarter $t - 1$ and $t - 2$.

The vector of covariates include *Experience* measured by the firm’s age, which captures network effects and advantageous relationships with debt providers a firm may build with time, deal entry-multiple (Enterprise Value/Sales), deal size and portfolio company revenue. Finally, I control for year-quarter, industry, investor country and portfolio-company country fixed effects. Apart from time fixed-effects, industry fixed-effects are particularly important since PE funds invest in correlated industries. In the absence of industry fixed-effects, industry-specific valuation changes could give rise to confounding effects.

I begin by running regressions in a manner similar to Jenkinson et al (2013). The main measure of performance that GPs and LPs discuss for private equity funds is the IRR. Any change in valuation will have an immediate effect on the reported IRR. Therefore, I first run the baseline specification using pre-acquisition quarterly change in the IRR, aggregated at the firm-quarter level using fund-size as weights. The results are reported in columns 1 to 4 of Table 2. I find a positive and statistically significant relationship across various specifications signalling that there may be valuation effects driving cross-border investments. However, as noted earlier, the IRR is heavily dependant on the timing of the cash flows into or out of a fund. Thus the primary message then from this set of regressions is that while quarterly change in IRR may signal valuation effects driving cross-border buyouts, they could also be confounded by cash flow effects.

I now turn to my primary measure of valuation outlined in equation 1. Results for the baseline specification, using quarterly change in NAV adjusted for calls and distributions, are summarized in Table 2, columns (5) to (8). Columns (5) to (8) deploy various portfolio-company and deal-specific controls¹². My main coefficient of interest, β_1 is positive and statistically significant. Column (5) presents results where I only control for portfolio company’s industry and country factors, not company-specific factors. When I introduce

¹²Note that I use a time-varying PE firm specific control, *Experience* which is omitted from the display for the purposes of brevity.

portfolio company revenue, Enterprise Value/Sales and deal size, the value of β_1 rises from 0.19 to 0.36 to 0.049.

To measure goodness of fit, I use McFadden’s *pseudo* – R^2 , which compared the model’s likelihood (L) to that of a model consisting of only a constant (L_0), i.e. the average cross-border buyout rate in the sample and can be interpreted as the standard R^2 . *Pseudo* – R^2 ranges from 0.47 to 0.54 in Table 2. Importantly, there is no difference in goodness of fit when I use change in cash-flow adjusted NAV compared to change in IRR suggesting valuation effects and not cashflow effects are driving cross-border buyout probabilities.

It is important to keep in mind that these results are cross-border relative to domestic buyout deals, providing evidence to the hypothesis that valuation effects are stronger in a cross-border context due to country-level macroeconomic factors such as real exchange rate and stock market movements. This finding is consistent with the broader cross-border merger and acquisitions literature seen in Baker et al (2009) and Erel et al (2012), but is a novel finding in the cross-border private equity literature.

5.3 Marginal Effects and Cross-border Buyouts

Ultimately, the economic significance of the coefficients from the logit specification lies in predicting actual cross-border buyouts. The disadvantage of utilizing odds ratios is that one cannot deduce a sense of magnitude. I seek to understand what the model is saying in the probability scale and not in the odds scale, much less in the estimation scale, the log-odds. This leads me to compute marginal effects which allow one to examine the impact of changes in a specific predictor variable on the probability of a cross-border buyout, holding all other predictors constant.

Similar to Asis, Chari and Haas (2020), in this section, I compute average marginal effects (AME_i) for one standard deviation of changes in the explanatory variables for each firm evaluation at their true valuation effects, experience and target characteristics across the sample. Next, I calculate marginal effects at the mean (MEM_i) - the effects of one standard

deviation change in the explanatory variable on the probability of cross-border buyouts for a representative firm with valuation effects, experience and target characteristics held at their respective sample means.¹³

The results are summarized in Table 3. Columns 1 to columns 3 present Average Marginal Effects while columns 4 to 6 present Marginal Effects at the Mean. The effects are actually quite large. AME for cash-flow adjusted change in NAV ranges from 0.001 to 0.004 in response to a one standard deviation shock to the covariate. Controlling for portfolio company revenue, column (5) reports MEM of 0.0123.

As Asis, Chari and Haas (2020) suggest, it is important to remember that although at first glance the magnitude of these effects may appear small, these are systemic effects and can translate to large numbers in predicted cross-border buyouts when aggregated across firms.

5.4 International Financial Conditions and Cross-Border Buyouts

Considering large share of debt in LBOs, a natural question concerns how cross-border buyouts are affected by global financial conditions. Axelson et al (2007) suggest that when debt markets periodically become "overvalued", investors do not demand the full interest rate corresponding to the fundamental underlying risk of a firm. This should lead firms to issue more debt when debt is more overvalued. In buyouts, if we presume LBO sponsors are rational, they would be expected to be willing to pay a higher price, and take on more leverage, when debt financing is "cheap" (Axelson et al (2007); Ljungqvist, Richardson and Wolfenzon (2007)). Asis, Chari and Haas (2020) suggest that a cost of external finance channel operates through different types of global funding shocks. Candidate global financial shocks that receive attention in the capital flows literature primarily falls into three inter-related categories: (i) U.S./advanced economy monetary policy shocks, (ii) shocks that impact foreign investor risk aversion and liquidity in international financial markets, and (iii) exchange rate shocks.

¹³See Asis, Chari and Haas (2020) for details on mathematical formalization of *AME* and *MEM*

Moreover, evidence suggests that these shocks characterize 'risk-on' and 'risk-off' states of the world. I suggest there is a differential response of cross-border buyout probabilities to valuation effects conditional on global financial conditions. Specifically, I expect favorable global financial conditions to strengthen the response of cross-border buyout probabilities to increasing valuation effects.

I begin with two proxies for U.S./advanced economy monetary policy shocks: First, I introduce the US Yield curve slope calculated as the difference between the 5-year treasury and the Fed funds rates. The Federal funds rate is indicative of monetary conditions and changes in monetary policy in the United States, whereas the slope of the yield curve captures how bond markets expect short-term interest rates (as a reflection of economic activity and future levels of inflation) to move in the future, and therefore borrowing costs at longer maturities. This is particularly applicable for leveraged buyouts which typically have holding periods of around 5 years (Bain and Company, 2019). Second, I introduce Excess Market Return calculated as the return on all NYSE, AMEX and NASDAQ stocks minus the one-month treasury bill rate. As Baker et al (2009) suggest, if equity valuations are high, cost of debt should be low and this effect will be transmitted to comparable private sector firms. To capture this monetary policy effect, I augment my baseline specification with interaction terms between cash-flow adjusted change in NAV and the quarterly Yield Curve Slope and Excess Market Return.

Results using these two interactions are summarized in Table 4. Columns (1) to (4) present estimates of the yield curve slope and valuation effects interaction and columns (5) to (8) present estimates of excess market return and valuation effects interaction. Consistent with my hypothesis, the interaction term between the yield curve slope and valuation effects is statistically significant and negative. This result is robust to alternate specifications using portfolio company controls, industry, country and year-quarter fixed effects. These results suggest as the yield curve becomes flatter, the effect of rising valuation effects on cross-border buyout probability becomes stronger. Looking at the estimates for the second interaction,

there is a positive and statistically significant relationship between the interaction term and cross-border buyout probabilities. This suggests as excess market return increases, the impact of rising valuation effects on cross-border buyout probabilities becomes larger. This finding is consistent with Baker et al (2009) who suggest that higher public equity valuations lead to lower cost of borrowing for private firms through widely used credit scoring models which uses share price as an input to determine credit risk of a private firm. It is worth noting that for all specifications in Table 4, the unconditional effect of cash-flow adjusted change in NAV remains positive and statistically significant at the 1 percent level, lending further credence to the primary hypothesis.

The next pair of global financial variables I introduce is the VIX and TED spread. Focusing on U.S. monetary policy, market risk and investor sentiments, Bekaert et al (2013), Miranda-Agrippino and Rey (2019) and many others use the VIX as a proxy to measure the risk appetite of global investors. Since the VIX captures the market's expectation for 30-day volatility in the S&P 500, a higher VIX typically denotes a general increase in risk premium, and consequently, an increase in borrowing costs. I suggest there is a correlation between cross-border buyout probabilities and a combination of valuation effects and foreign investor risk appetites. Fratzscher (2012) includes the TED spread, calculated as the spread between 3-month LIBOR Rates and 3-month T-bill rates, to capture credit risk and liquidity in international capital markets. To capture how valuation effects on cross-border buyout probabilities are shaped by these variables, I introduce interactions between cash-flow adjusted NAV change and the VIX and TED spread.

Table 5 presents findings from this specification. Without accounting for deal-specific or portfolio company accounting variables, I do not find a statistically significant effect between the TED spread and valuation effects interaction and cross-border buyout probabilities. However, as column (2) shows, introducing portfolio company revenue results in a negative and statistically significant coefficient at the 5 percent significance level on the interaction term. When I introduce EV/Sales and deal value, the statistical significance rises to the

1 percent level. Similar patterns emerge for the interaction between VIX and change in $NAVCF$. For all specifications in columns (5) to (8), the interaction term is negative and statistically significant at the 1 percent level. Moreover, the unconditional effect of the primary variable of interest, cash-flow adjusted NAV change remains positive and statistically significant at the 1 percent level. These results provide evidence to the hypothesis that favorable global liquidity conditions and an increase in foreign investors' appetite for risk, captured by falling TED Spread and VIX, strengthens the positive association between PE firm valuations and cross-border buyout probabilities.

Finally, to further characterize how cross-border buyouts and valuation effects are tied to "risk-on", "risk-off" states of the world, I introduce two more variables: first, I construct a RORO index. Despite being somewhat imprecisely defined, the RORO terminology has come into pervasive use in the financial press and among policy makers in the years since the global financial crisis. Second I introduce the change in the broad dollar index, calculated using trade-weighted averages of exchange rates from a broad basket of US trading partners; the broad dollar is said to appreciate as the index value rises. For both variables, an increase in value indicates a 'risk-off' state of the world, and *vice versa*. Problems of information asymmetry are particularly severe in cross-border deals due to unfamiliar legal regimes, accounting standards, language or corporate cultures, inherently making cross-border deals risky relative to domestic deals. Interactions with $\Delta NAVCF$ show if the positive relationship between cross-border buyouts and valuation effects is stronger in the face of favorable, risk-on shocks.

Table 6 reports this set of results. Columns (1) to (4) presents results with interactions of the RORO index, columns (5) to (8) with the broad dollar index. For all specifications using the RORO index, I find a statistically significant and negative relationship between the interaction term and cross-border buyout probabilities. When I control for deal size and target revenue (column 2) or entry multiple (column 3), the significance level rises from 5 percent to the 1 percent level. Moving on to the change in broad dollar index, the results

echo a similar pattern: the interaction term has a statistically significant and negative effect at the 1 percent level. These results indicate that a favorable, 'risk-on' shock strengthens the impact of valuation effects on cross-border buyout probabilities.

5.5 PE Firm Heterogeneity and Cross-Border Buyouts

My primary results indicate that PE firm valuation has a material impact on the probability of a cross-border buyout by reducing information asymmetry costs. Reputation is an important part of the story since it provides lenders valuable information about PE firms on their ability to make operational improvements to portfolio companies. In this section, I investigate how differences in PE firm reputation affect the relationship between cross-border buyout probability and valuations. I conjecture that the impact of valuation changes on cross-border buyout probabilities is stronger for low reputation PE firms with greater information asymmetry costs relative to high reputation PE firms. I extend model (i) by including the interaction of $\Delta NAVCF$ with a $Reputation_j$ dummy which is defined in a number of ways in terms of performance, frequency of interaction with lending markets and firm size, consistent with the literature:

- $Reputation_{it}$ takes value of 0 for high reputation and 1 for low reputation; a firm is defined as high reputation in a given firm-quarter if its NAV is at the top-quartile of historical NAVs sorted by PE firm. Low reputation firms are the complements to high reputation firms.
- $Reputation_{it}$ takes value of 0 for high reputation and 1 for low reputation; a firm is defined as high reputation in a given firm-quarter if its NAV is at the top-tercile of historical NAVs sorted by PE firm. Low reputation firms are the complements to high reputation firms.
- $Reputation_{it}$ takes value of 0 for high reputation and 1 for low reputation; a firm is defined as low reputation in a given firm-quarter if its Investment Multiple is below

the median historical Investment Multiple. High reputation firms are the complements to low reputation firms.

- $Reputation_{it}$ takes value of 0 for high reputation and 1 for low reputation; a firm is defined as high reputation in a given firm-quarter if its total number of buyout deals in the sample is at the top-quartile of number of buyout deals sorted by PE firm. Low reputation firms are the complements to high reputation firms.
- $Reputation_{it}$ takes value of 0 for high reputation and 1 for low reputation; a firm is defined as high reputation in a given firm-year if its capital commitments is at the top-decile of sample capital commitments sorted by PE firm. Capital commitments is the total value of all funds managed by a PE firm and represents firm size. Low reputation firms are the complements to high reputation firms.
- $Reputation_{it}$ takes value of 0 for high reputation and 1 for low reputation; a firm is defined as high reputation in a given firm-quarter if its capital commitments is above \$ 20 billion. Low reputation firms are the complements to high reputation firms.

The first two variables, NAV and the Investment Multiple capture fund performance. To the extent that NAV captures the ability of PE firms to make operational improvements on their portfolio companies, we can use this variable to gauge reputation heterogeneity across firm-quarters. Similarly, the Investment Multiple or Total Value/Paid-in Capital captures the ability to add value to portfolio companies, adjusting for the book value of the investment and is commonly used in the literature. The advantage of these measures is that they are time-varying allowing lenders to update their perception of the sponsoring firm based on recent performance.

The reduced information asymmetry costs argument also emanates from the repeated, highly frequent interactions between PE investors and banks as compared to traditional corporate borrowers (Ivanisha and Kovner, 2010). Consistent with this idea, I compute 'number of total buyout transactions' for each PE firm to capture frequency of interaction

with lending markets. Another prominent measure of information asymmetry costs (or reputation) is size of the PE group, which I capture through capital commitments similar to Achleitner, Braun, Hinterramskogler and Tappeiner (2012) and Demiroglu and James (2010). Figure 5 in the Appendix plots number of firm-quarters across bins for Assets Under Management. There is a sharp drop in number of firm-quarters around the USD 20 billion mark, motivating the final definition.

Table 7 reports estimates where $Reputation_j$ is modelled using NAV and Table 8 reports estimates where $Reputation_j$ is modelled using Total Value/Paid-in Capital. Focusing on Table 7 columns (1) and (2), the coefficient of the interaction term is positive and statistically significant at the 5 percent level suggesting valuation effects are stronger for low reputation firms. When I control for entry multiple or deal size the statistical significance rises to 1 percent. For three of the four specifications in columns (1) to (4), the unconditional effect of the cash-flow adjusted NAV change remains positive and statistically significant. The estimates using top tercile NAVs echo a similar pattern¹⁴. Next turning to table 8, first I now note that the unconditional effect of the Reputation dummy is statistically significant and negative. This is not surprising since high reputation firms are likely to have greater access to the cross-border buyout market. Consistent with the findings in Table 7, the coefficient of the interaction term is again positive and statistically significant. For specifications (2) and (3) the unconditional valuation effects also remains positive and statistically significant.

Focusing on Table 9, reputation is now modelled as the frequency of interaction with lending markets. Since a buyout deal is characterized by bank debt, number of completed deals by a PE firm captures the frequency of interaction between the PE firm and banks. Defining reputation in this manner, I find that the coefficient of the interaction term is again positive and statistically significant at the 1 percent level. Note also the unconditional effect of the reputation dummy is negative and statistically significant at the 1 percent level

¹⁴The unconditional effect on $Reputation_j$ is positive and statistically significant at the 10 percent level in column 1. As before, this specification does not control for portfolio company or deal characteristics. Once I control for portfolio company revenue or deal size the statistical significance disappears. Caution is therefore warranted in making interpretations on this coefficient.

consistent with Table 8. These results indicate that the effect of a positive valuation shock on cross-border buyout probabilities is stronger for low reputation firms.

At first pass this might seem counter-intuitive since high reputation firms such as KKR or Blackstone are much more active in the cross-border LBO market. However considering that heterogeneity in reputation captures information asymmetry cost differences, the impact of an upward valuation shock on information asymmetry costs is greater for low reputation firms. Consider a simple example between two PE firms: a high reputation firm such as KKR and a low reputation firm. A positive valuation shock will benefit the low reputation firm relatively more, since lenders will be already willing to lend to KKR given its reputation capital and associated low asymmetric information costs. As a robustness check, I also use PE firm size as a measure of reputation. Table A4 in the Appendix reports these results. The results are qualitatively similar.

6 Conclusion

There is a dearth of research looking at the role valuations effects play on cross-border buyout investments. Valuation changes can be the result of fund-specific or market-wide shocks such as exchange rate movements or stock market fluctuations. Several studies show that higher private equity groups reduce asymmetric information costs leading to more favorable borrowing conditions through their general reputation and past performance. Consistent with this idea, I conjecture that cashflow-adjusted valuation measures are positively associated with the probability of a cross-border buyout. I disentangle valuation effects from a private equity fund's NAV, and show that private equity firm valuations are positively related to the probability of a cross-border buyout, relative to a domestic buyout. This result is robust to PE firm, target firm, country and external market conditions. Introducing marginal effects shows that the marginal change in probability of a cross-border buyout in response to a valuation change is non-trivial.

I extend my baseline results in two main directions: external market conditions and PE firm heterogeneity. First, since the hallmark of a buyout transaction is high levels of debt, I explore how changes in global financial conditions affect my baseline hypothesis. I find that easing monetary conditions in U.S./advanced economies, falling VIX, TED spread and broad dollar index strengthens the effect valuation changes on cross-border buyout probability. More generally, the results indicate that a 'risk-on' shock to global funding conditions increases the positive impact of valuation effects on cross-border buyout probabilities. This finding is novel. Second, the paper explores how heterogeneity in PE firm reputation governs the relationship between valuation and cross-border buyouts. Modelling reputation in terms of performance, firm size and frequency of interaction with lending markets, I find that valuation effects are particularly strong for low reputation PE firms facing relatively higher agency cost of lending. This finding is novel in the context of private equity. Overall my results contribute to the literature on investment behavior of private equity buyout funds.

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Figures

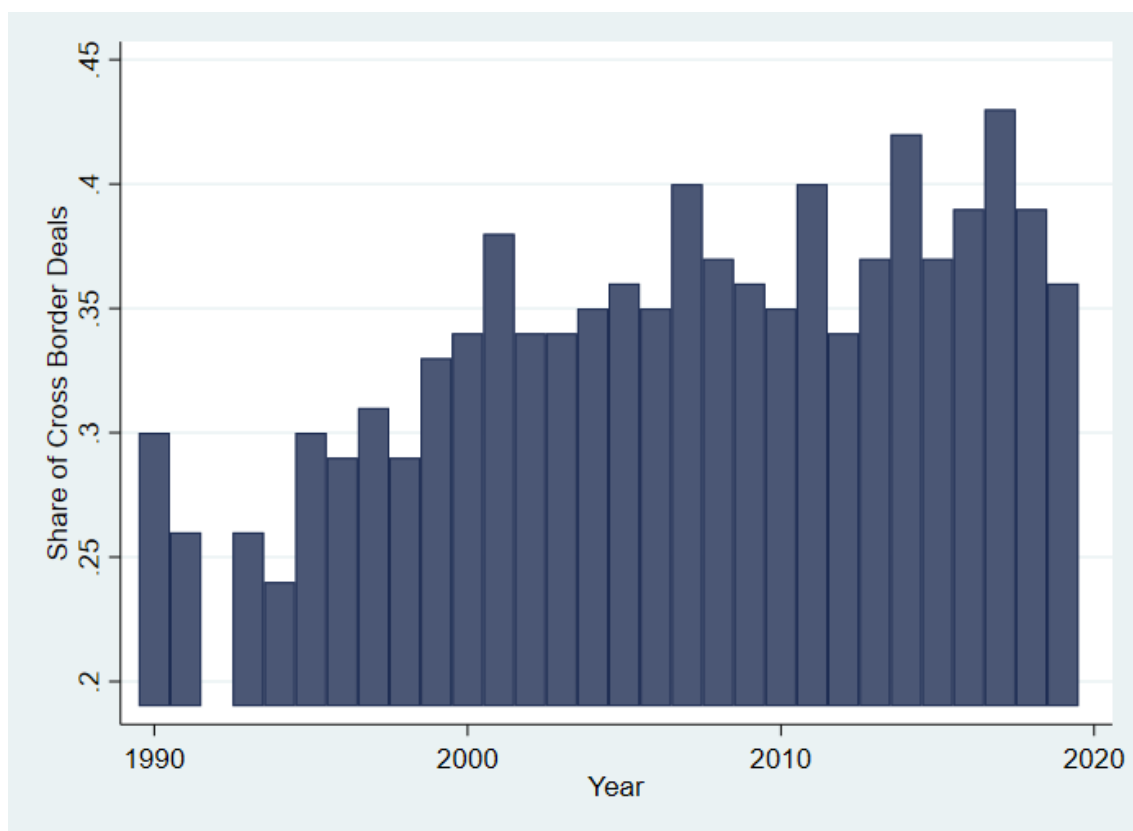


Figure 1: Share of Cross-Border Buyout Deals (Deal Count)

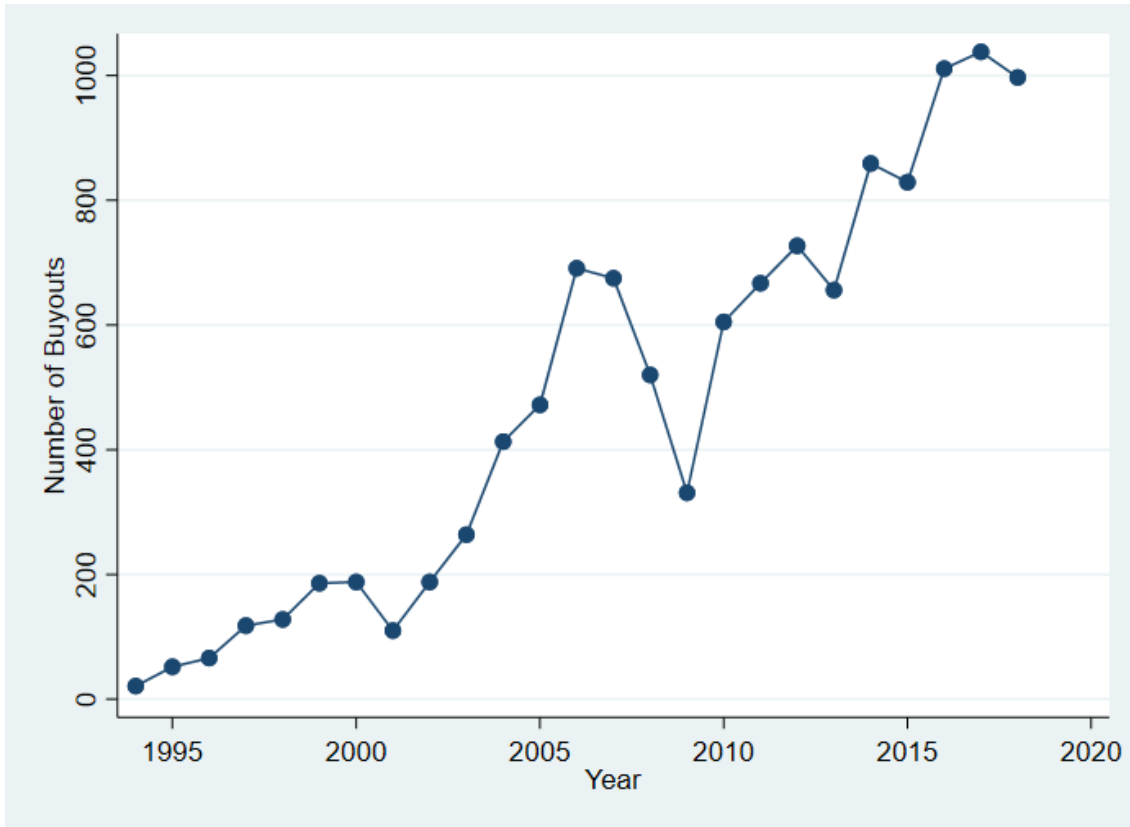


Figure 2: Number of Buyout Transactions

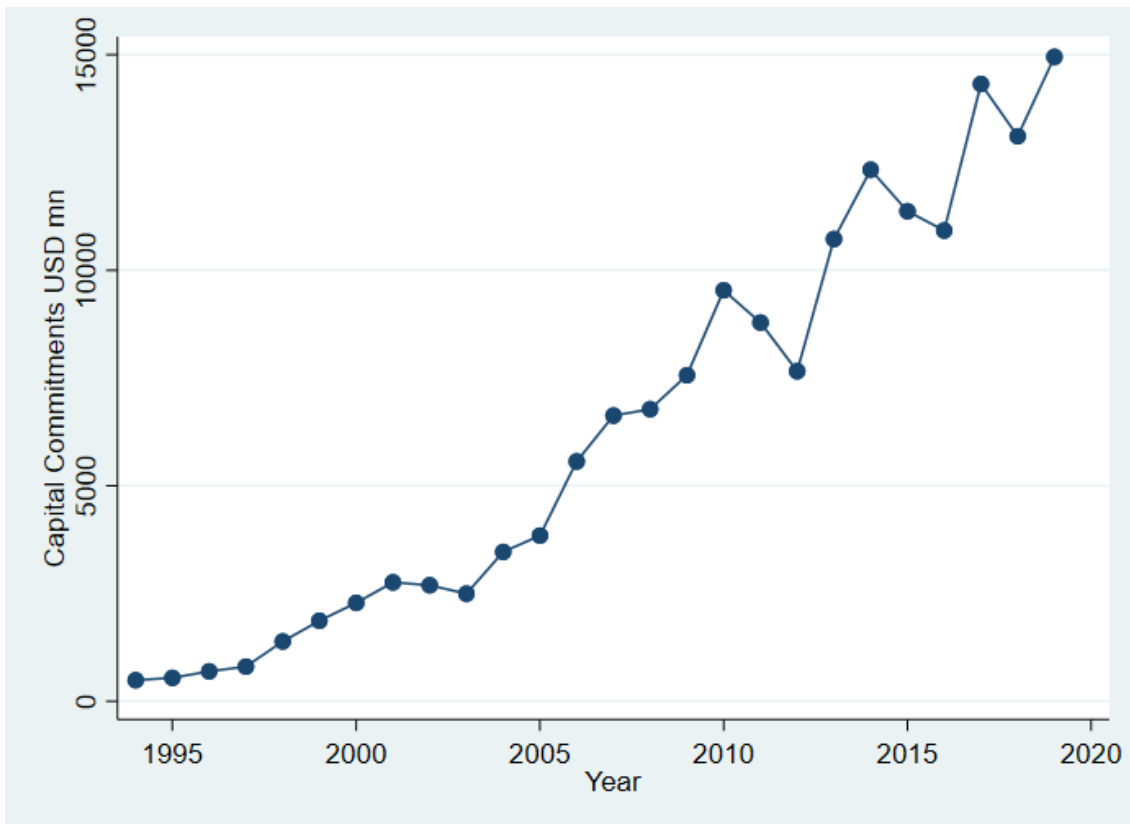


Figure 3: Global Buyout Committed Capital

Tables

Table 1: Summary Statistics

Summary Statistics for all private equity fund, private equity firm-level and portfolio-company variables. The first two rows present unmatched full sample from Preqin. The third row presents the sum of all funds managed by a PE firm, or total Firm Capital Commitments. Variables from Row 4-7 are weighted means, with fund size as weights. Firm Age is the age of the firm at the time of a buyout transaction.

	Obs	Mean	Std. Dev	p25	p50	p75	p95
Panel A: Private Equity firm							
Fund IRR (Full Preqin sample)	81154	11.16	25.39	0.90	8.83	17.44	40.47
Fund Size (Full Preqin sample, \$ Mn)	213715	992	1959	175	389	890	3873
Firm Capital Commitments (\$ Mn)	11344	8721	14641	725	2275	8426	46962
Fund-Size Weighted Firm IRR	11075	11.83	13.62	4.1	11.38	17.45	35.53
Fund-Size Weighted Δ Firm IRR	11429	0.33	2.01	-0.11	0.00	0.58	3.30
Fund-Size Weighted Firm NAV (\$ Mn)	11551	4.85	2.40	3.30	4.80	6.20	8.90
Fund-Size weighted Firm Δ NAVCF	11837	2.33	5.18	0.00	1.51	3.54	9.58
Fund-Size weighted Firm Investment Multiple	11398	1.30	0.42	1.04	1.29	1.52	2.35
Firm Age	11608	21.40	12.04	13.00	20.00	28.00	46.00
Panel B: Portfolio Company							
Revenue (\$ mn)	2901	5457	72612	74	250	816	5900
Entry Multiple (Enterprise Value/Sales)	2001	3.97	40.32	0.4	1.48	2.69	6.41
Deal Value (\$ mn)	4390	990	2535	100	306	909	3600

Table 2: **Logit Regressions of Probability of Cross-Border Buyouts next quarter**

This table summarizes results of the logit regressions where the dependant variable takes a value of 1 if a buyout deal is cross-border and 0 otherwise. Explanatory variables include cash-flow adjusted quarterly change in Net Asset Value, $\Delta NAVCF$, of a private equity fund, aggregated at the firm-quarter level using fund-size weighted average, quarterly change in IRR, ΔIRR , of a private equity fund, aggregated at the firm-quarter level using fund-size weighted average, PE firm age, Portfolio Company Revenue, Enterprise Value/Sales and Deal Value. All specifications control for PE firm country (acquirer), portfolio firm country (target), portfolio firm's primary industry and Year-quarter fixed effects. Standard errors are clustered at the country-pair level.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
ΔIRR	0.0485*** (0.0112)	0.0706** (0.0302)	0.0586*** (0.0128)	0.0515** (0.0213)				
$\Delta NAVCF$					0.0192*** (0.00508)	0.0498*** (0.0106)	0.0369*** (0.00619)	0.0449*** (0.0123)
Target_Revenue		0.215*** (0.0444)				0.222*** (0.0454)		
Deal_Size			0.0000433*** (0.0000113)				0.0000448*** (0.0000109)	
Entry_Multiple				0.000310 (0.000681)				0.0000234 (0.000707)
Pseudo R-squared	0.544	0.521	0.540	0.491	0.543	0.527	0.544	0.471
N	10868	2488	3795	1545	10934	2498	3856	1508
Country FE	Y	Y	Y	Y	Y	Y	Y	Y
Industry FE	Y	Y	Y	Y	Y	Y	Y	Y
Year-Quarter FE	Y	Y	Y	Y	Y	Y	Y	Y

Standard errors in parentheses

* $p < .10$, ** $p < .05$, *** $p < .01$

Table 3: **Marginal Effects**

This table summarizes Average Marginal Effects (AME) and Marginal Effects at the Mean (MEM) of the logit regressions where the dependant variable takes a value of 1 if a buyout deal is cross-border and 0 otherwise. Columns (1) to (3) present AME and columns (4) to (6) present MEM. Explanatory variables include cash-flow adjusted quarterly change in Net Asset Value, $\Delta NAVCF$, of a private equity fund, aggregated at the firm-quarter level using fund-size weighted average, PE firm age, Portfolio Company Revenue, Enterprise Value/Sales and Deal Value. All specifications control for PE firm country (acquirer), portfolio firm country (target), portfolio firm's primary industry and Year-quarter fixed effects. Standard errors are clustered at the country-pair level.

	(1)	(2)	(3)	(4)	(5)	(6)
$\Delta NAVCF$	0.00156** (0.000726)	0.00486*** (0.00159)	0.00331* (0.00173)	0.00347** (0.00149)	0.0123*** (0.00244)	0.00864*** (0.00225)
Target_Revenue		0.0217** (0.0110)			0.0551*** (0.0120)	
Deal_Size			0.00000403* (0.00000217)			0.0000105*** (0.00000335)
N	10934	2498	3856	10934	2498	3856
Country FE	Y	Y	Y	Y	Y	Y
Industry FE	Y	Y	Y	Y	Y	Y
Year-Quarter FE	Y	Y	Y	Y	Y	Y

Standard errors in parentheses

* $p < .10$, ** $p < .05$, *** $p < .01$

Table 4: **International Financial Conditions, Valuation Effects and Probability of Cross-border Buyouts**

This table summarizes results of the logit regressions where the dependant variable takes a value of 1 if a buyout deal is cross-border and 0 otherwise. Explanatory variables include all variables used in the baseline specification. In addition, it includes Yield curve slope (Yld Slope) and Excess Market return. Yield curve slope is calculated as the difference between the US 5-year Treasury rate and the Fed Funds rate. Excess Market return (Excess Return) is calculated as the value-weight return on all NYSE, AMEX, and NASDAQ stocks (from CRSP) minus the one-month Treasury bill rate. Finally, there are two interaction terms between Yield curve slope and cash-flow adjusted NAV change and Excess Market return and cash-flow adjusted NAV change. All specifications control for PE firm country (acquirer), portfolio firm country (target), portfolio firm's primary industry and Year fixed effects. Standard errors are clustered at the country-pair level.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta NAVCF$	0.0236*** (0.00716)	0.0755*** (0.0103)	0.0601*** (0.00695)	0.0723*** (0.0124)	0.0357*** (0.00568)	0.0396*** (0.0109)	0.0386*** (0.0110)	0.0389*** (0.0110)
$\Delta NAVCF * YldSlope$	-0.0740** (0.00646)	-0.0330*** (0.00835)	-0.0290*** (0.00800)	-0.0316*** (0.0114)				
$\Delta NAVCF * ExcessReturn$					0.00597** (0.00237)	0.0179*** (0.00373)	0.0184*** (0.00395)	0.0184*** (0.00392)
Target_Revenue		0.222*** (0.0458)				0.172*** (0.0507)		
Deal_Size			0.0000444*** (0.0000111)		0.0000457*** (0.0000112)	0.00000240 (0.0000145)		0.0000363** (0.0000163)
Entry_Multiple				0.000178 (0.000720)			0.0000267 (0.000740)	-0.0000961 (0.000770)
Excess_Return					0.00996 (0.0328)	0.0174 (0.0418)	0.00442 (0.0411)	0.00989 (0.0427)
Yld Slope	-5.219 (5.797)	-5.702 (5.110)	-3.546 (5.856)	-5.961 (8.198)				
Pseudo R-squared	0.543	0.528	0.544	0.472	0.544	0.477	0.473	0.474
N	10934	2498	3856	1508	3856	1508	1508	1508
Country FE	Y	Y	Y	Y	Y	Y	Y	Y
Industry FE	Y	Y	Y	Y	Y	Y	Y	Y
Year-Quarter FE	Y	Y	Y	Y	Y	Y	Y	Y

Standard errors in parentheses

* $p < .10$, ** $p < .05$, *** $p < .01$

Table 5: **Global Investors' Risk Appetite, Liquidity Shocks and Probability of Cross-border Buyouts**

This table summarizes results of the logit regressions where the dependant variable takes a value of 1 if a buyout deal is cross-border and 0 otherwise. Explanatory variables include all variables used in the baseline specification. In addition, it includes the Chicago Board Options Exchange's CBOE Volatility Index (VIX) and TED spread (TED) calculated as the spread between 3-month Libor based on U.S. Dollars and 3-month Treasury Bill Rate. Finally, there is an interaction term between $\Delta NAVCF$ and the two previous variables mentioned. All specifications control for PE firm country (acquirer), portfolio firm country (target), portfolio firm's primary industry and Year fixed effects. Standard errors are clustered at the country-pair level.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta NAVCF$	0.0282*** (0.0107)	0.0681*** (0.0125)	0.0872*** (0.0188)	0.0858*** (0.0183)	0.132*** (0.0230)	0.174*** (0.0242)	0.173*** (0.0243)	0.173*** (0.0243)
$\Delta NAVCF_TED$	-0.0159 (0.0162)	-0.0311** (0.0126)	-0.0713*** (0.0266)	-0.0701*** (0.0250)				
$\Delta NAVCF_VIX$					-0.00439*** (0.00147)	-0.00696*** (0.00116)	-0.00694*** (0.00115)	-0.00692*** (0.00116)
Target_Revenue		0.221*** (0.0458)	0.173*** (0.0505)		0.221*** (0.0455)	0.172*** (0.0504)		
Deal_Size			0.000000405 (0.0000142)			0.000000108 (0.0000148)		0.0000347** (0.0000163)
Entry_Multiple				-0.000159 (0.000716)			0.0000749 (0.000726)	-0.0000334 (0.000754)
TED	-17.45 (19.42)	-19.34 (17.61)	-15.85 (20.40)	-20.01 (27.44)				
VIX					-1.389 (1.262)	-1.143 (1.496)	-1.443 (2.002)	-1.431 (1.950)
Pseudo R-squared	0.543	0.527	0.476	0.472	0.528	0.478	0.474	0.474
N	10934	2498	1508	1508	2498	1508	1508	1508
Country FE	Y	Y	Y	Y	Y	Y	Y	Y
Industry FE	Y	Y	Y	Y	Y	Y	Y	Y
Year-Quarter FE	Y	Y	Y	Y	Y	Y	Y	Y

Standard errors in parentheses

* $p < .10$, ** $p < .05$, *** $p < .01$

Table 6: **Risk-on/Risk-off, Valuation Effects and Probability of Cross-border Buyouts**

This table summarizes results of the logit regressions where the dependant variable takes a value of 1 if a buyout deal is cross-border and 0 otherwise. Explanatory variables include all variables used in the baseline specification. In addition, it includes a Risk-on,Risk-off (RORO) Index and an interaction term between the same and *NAVCF* in columns (1) to (4). In columns (5) to (8), it includes change in the Broad Dollar defined as Real Trade Weighted U.S. Dollar Index. All specifications in columns (1) to (4) control for country, industry and year-quarter fixed effects. Country fixed effects are omitted in columns (5) to (8). Standard errors are clustered at the country-pair level.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta NAVCF$	0.0129*** (0.00487)	0.0257 (0.0167)	0.0238 (0.0170)	0.0296*** (0.00546)	0.00535 (0.00715)	0.0223 (0.0155)	0.0158 (0.0116)	0.0264 (0.0180)
RORO* $\Delta NAVCF$	-0.903** (0.442)	-2.446*** (0.905)	-2.526** (0.981)	-1.125** (0.441)				
$\Delta BroadDollar$ * $\Delta NAVCF$					-0.0205*** (0.00744)	-0.0209** (0.00966)	-0.0213*** (0.00684)	-0.0236*** (0.00880)
RORO	62.96 (68.17)	58.52 (67.79)	73.55 (89.59)	42.59 (65.71)				
Target_Revenue		0.171*** (0.0494)				0.193 (0.171)		
Deal_Size		0.00000235 (0.0000141)		0.0000449*** (0.0000110)		-0.000132** (0.0000523)	-0.0000480 (0.0000480)	
Entry_Multiple			-0.0000530 (0.000711)					-0.000372 (0.00143)
$\Delta BroadDollar$					0.943 (8.098)	-3.200 (9.685)	0.222 (7.231)	-7.273 (7.666)
Pseudo R-squared	0.543	0.475	0.471	0.544	0.0270	0.103	0.0558	0.0841
N	10934	1508	1508	3856	9625	1677	3425	1677
Country FE	Y	Y	Y	Y	N	N	N	N
Industry FE	Y	Y	Y	Y	Y	Y	Y	Y
Year-Quarter FE	Y	Y	Y	Y	Y	Y	Y	Y

Standard errors in parentheses

* $p < .10$, ** $p < .05$, *** $p < .01$

Table 7: **Reputation Effects by Top Quartile and Top Tercile NAVs**

This table summarizes results of the logit regressions where the dependant variable takes a value of 1 if a buyout deal is cross-border and 0 otherwise. Explanatory variables include all variables used in the baseline specification. In addition, it has a variable called Reputation which takes the value of 0 for high reputation and 1 for low reputation; in columns (1)-(4), a firm is defined as a high reputation firm in a given firm-quarter if its NAV is at the top-quartile of sample NAVs; in columns (5)-(8), a firm is defined as a high reputation firm in a given firm-quarter if its NAV is at the top-tercile of sample NAVs;. Finally it includes an interaction between Reputation and $\Delta NAVCF$. All specifications control for PE firm country (acquirer), portfolio firm country (target), portfolio firm's primary industry and Year fixed effects. Standard errors are clustered at the country-pair level.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta NAVCF$	0.0225** (0.0111)	0.0451*** (0.0162)	0.0234** (0.0119)	0.0136 (0.0154)	0.0204** (0.0104)	0.0463*** (0.0146)	0.0244** (0.0108)	0.00999 (0.0124)
Reputation* $\Delta NAVCF$	0.0172** (0.008)	0.0418** (0.0210)	0.0402*** (0.0132)	0.0663*** (0.0205)	0.0156 (0.0105)	0.0378* (0.0203)	0.0408*** (0.0130)	0.0832*** (0.0227)
Reputation	0.146* (0.0823)	0.0172 (0.120)	-0.0256 (0.113)	-0.0863 (0.142)	0.0191 (0.0677)	-0.0197 (0.105)	-0.0551 (0.0860)	-0.118 (0.151)
Target_Revenue		0.226*** (0.0419)				0.227*** (0.0421)		
Deal_Size			0.0000413*** (0.0000112)				0.0000416*** (0.0000112)	
Entry_Multiple				-0.000317 (0.000723)				-0.000313 (0.000757)
Pseudo R-squared	0.541	0.534	0.552	0.493	0.540	0.534	0.552	0.493
N	10464	2400	3743	1462	10464	2400	3743	1462
Country FE	Y	Y	Y	Y	Y	Y	Y	Y
Industry FE	Y	Y	Y	Y	Y	Y	Y	Y
Year-Quarter FE	Y	Y	Y	Y	Y	Y	Y	Y

Standard errors in parentheses

* $p < .10$, ** $p < .05$, *** $p < .01$

Table 8: **Reputation Effects by Investment Multiples**

This table summarizes results of the logit regressions where the dependant variable takes a value of 1 if a buyout deal is cross-border and 0 otherwise. Explanatory variables include all variables used in the baseline specification. In addition, it has a dummy called Reputation which takes the value of 0 for high reputation and 1 for low reputation; a firm is defined as a low reputation firm in a given firm-quarter if its Investment Multiple, defined as Total Value/Paid-in Capital, is below the median historical Total Value/Paid-in Capital. Finally it includes an interaction between Reputation and $\Delta NAVCF$. All specifications control for PE firm country (acquirer), portfolio firm country (target), portfolio firm's primary industry and Year fixed effects. Standard errors are clustered at the country-pair level.

	(1)	(2)	(3)	(4)
$\Delta NAVCF$	0.00212 (0.0110)	0.0284* (0.0156)	0.0156** (0.0052)	0.0120 (0.0180)
Reputation* $\Delta NAVCF$	0.0396** (0.0185)	0.0567** (0.0251)	0.0409** (0.0196)	0.0650*** (0.0251)
Reputation	-0.479*** (0.125)	-0.641*** (0.0912)	-0.379** (0.152)	-0.623*** (0.165)
Target_Revenue		0.165*** (0.0450)		
Deal_Size			0.0000401*** (0.0000112)	
Entry_Multiple				-0.000161 (0.000673)
Pseudo R-squared	0.541	0.537	0.547	0.484
N	10054	2329	3613	1426
Country FE	Y	Y	Y	Y
Industry FE	Y	Y	Y	Y
Year-Quarter FE	Y	Y	Y	Y

Standard errors in parentheses

* $p < .10$, ** $p < .05$, *** $p < .01$

Table 9: **Reputation Effects through Repeated Interaction with Lending Market**

This table summarizes results of the logit regressions where the dependant variable takes a value of 1 if a buyout deal is cross-border and 0 otherwise. Explanatory variables include all variables used in the baseline specification. In addition, it has a dummy called Reputation which takes the value of 0 for high reputation and 1 for low reputation; a firm is defined as a high reputation firm its total number of buyout deals is at the top-quartile of number of buyout deals sorted by PE firm. Finally it includes an interaction between Reputation and $\Delta NAVCF$. All specifications control for PE firm country (acquirer), portfolio firm country (target), portfolio firm's primary industry and Year fixed effects. Standard errors are clustered at the country-pair level.

	(1)	(2)	(3)	(4)
$\Delta NAVCF$	0.00187 (0.00514)	-0.0108 (0.0239)	0.0111 (0.0196)	0.0124 (0.0184)
Reputation* $\Delta NAVCF$	0.0313*** (0.0111)	0.0791** (0.0312)	0.0717** (0.0333)	0.0721** (0.0321)
Reputation	-1.165*** (0.176)	-1.345*** (0.182)	-1.191*** (0.247)	-1.253*** (0.263)
Target_Revenue		0.147*** (0.0383)	0.116*** (0.0392)	
Deal_Size			-0.00000603 (0.0000153)	
Entry_Multiple				-0.000741 (0.000758)
Pseudo R-squared	0.559	0.544	0.490	0.488
N	10934	2498	1508	1508
Country FE	Y	Y	Y	Y
Industry FE	Y	Y	Y	Y
Year-Quarter FE	Y	Y	Y	Y

Standard errors in parentheses

* $p < .10$, ** $p < .05$, *** $p < .01$

Appendix: Cross-Border Private Equity and Valuation Effects

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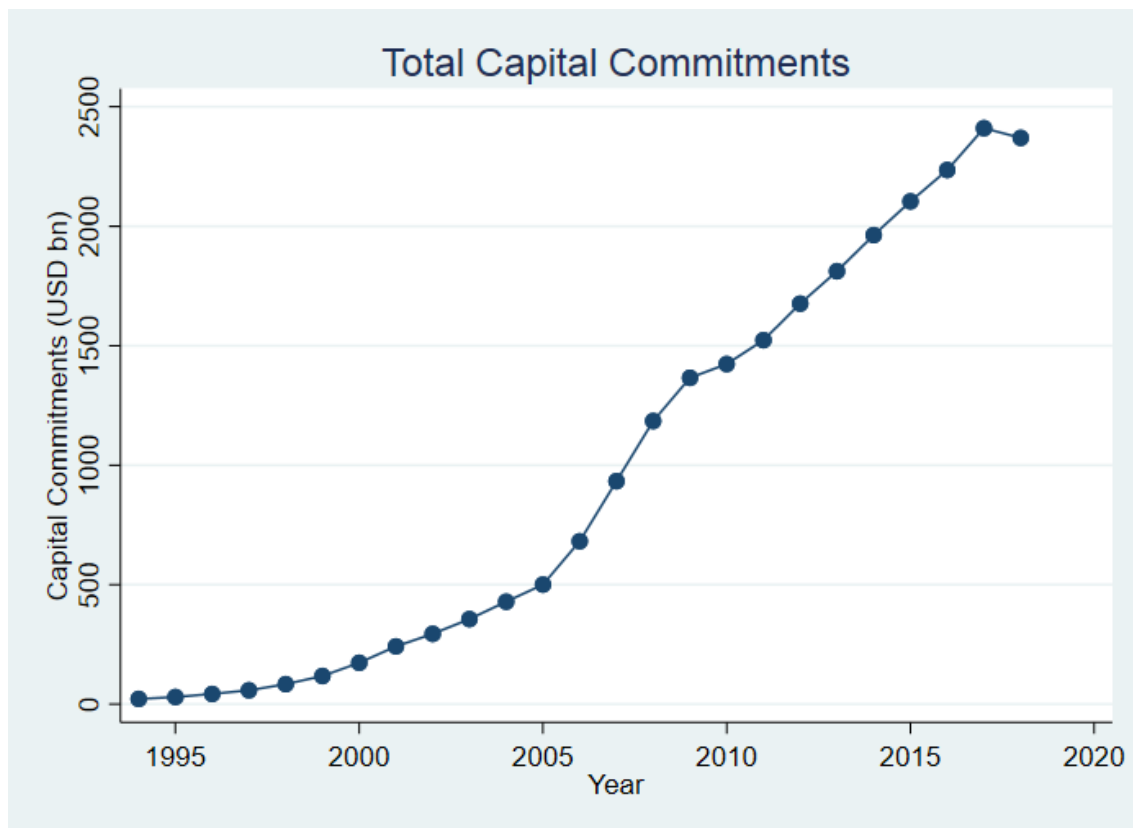


Figure A1: Global PE Capital Commitments: Full Preqin Sample Notes: This chart aggregates capital commitments to all types of private equity funds including Buyout, Venture capital, Mezzanine, Growth Capital, Recapitalization and Turnarounds

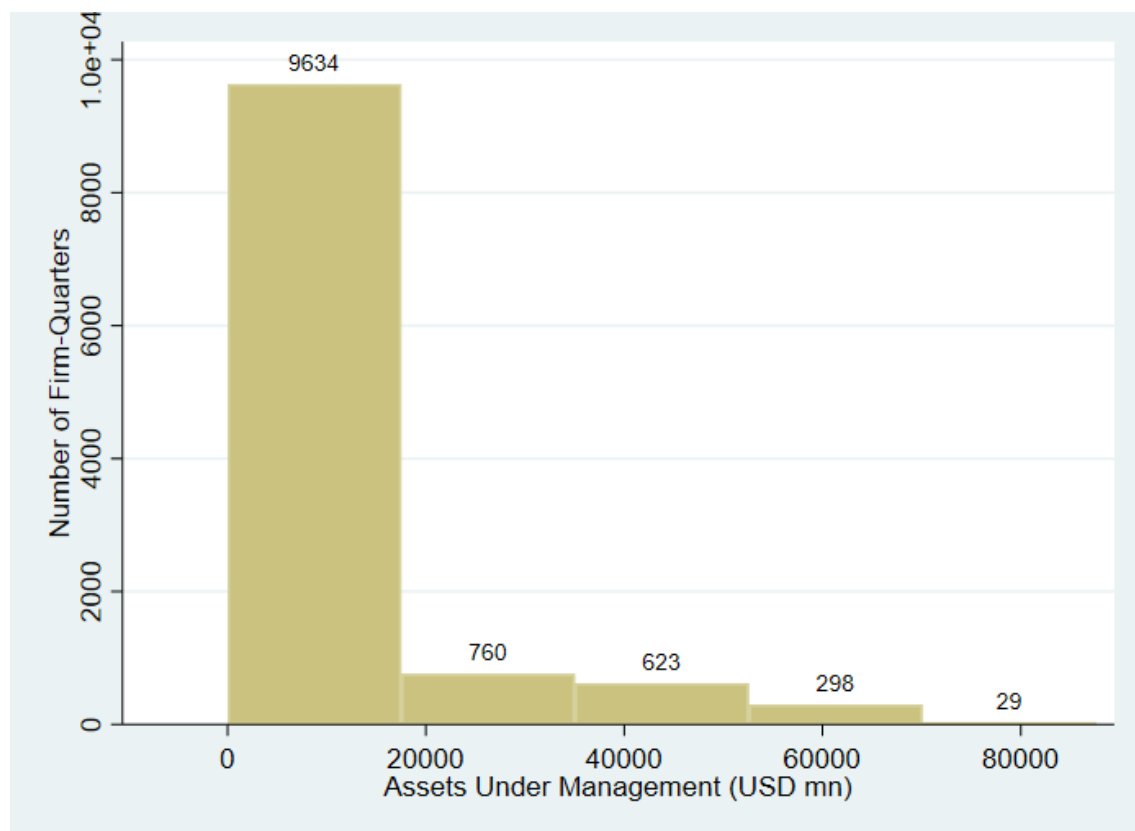


Figure A2: Firm Size Distribution: Assets Under Management

Table A1: Top Investor Countries by Buyout count

Investor Country	Total Deals	Total Cross-border deals	Share of Cross-Border deals
US	7637	1772	23.2
UK	1851	1032	55.8
France	356	152	42.7
Netherlands	173	79	45.7
Sweden	167	113	67.7
Canada	106	70	66.0
Switzerland	101	93	92.1
Norway	91	53	58.2
Finland	78	34	43.6
Australia	71	7	9.9
Italy	70	8	11.4
Denmark	42	9	21.4
South Africa	39	5	12.8
Japan	37	7	18.9
China	32	17	53.1
Total Buyouts	10851	3451	31.8

Table A2: Top Target Countries by Buyout count

Investor Country	Total Deals	Total Cross-border deals	Share of Cross-Border deals
US	6200	335	5.40
UK	1231	412	33.47
France	485	281	57.94
Germany	419	160	38.19
Netherlands	254	84	33.07
Italy	239	163	68.20
Sweden	217	72	33.18
Canada	183	147	80.33
Spain	178	37	20.79
Australia	156	92	58.97
India	133	24	18.05
China	133	118	88.72
Denmark	122	89	72.95
Norway	110	34	31.10
Brazil	102	26	25.00
Total Buyouts	10162	2074	20.4

Table A3: Summary Statistics: Global Financial Variables

	Obs	Mean	Std. Dev	p25	p50	p75	p95
Fed Funds Rate	11837	1.63	1.9	0.14	0.79	2.27	5.27
5-year Treasury	11837	2.59	1.48	1.52	2.07	3.5	5.55
VIX	11837	17.65	6.25	13.407	16.072	20.67	29.93
TED Spread	11837	0.42	0.31	0.229	0.309	0.495	1.12
Excess Market Return	11837	0.686	2.25	0.001367	0.976	1.9367	3.9367
Delta Broad Dollar	11837	0.05	0.77	-0.5	0	0.51	1.55

Table A4: **Reputation Effects by Capital Commitments**

This table summarizes results of the logit regressions where the dependant variable takes a value of 1 if a buyout deal is cross-border and 0 otherwise. Explanatory variables include all variables used in the baseline specification. In addition, it has a variable called Reputation which takes the value of 0 for high reputation and 1 for low reputation; in columns (1)-(3), a firm is defined as a high reputation firm in a given firm-quarter if has top-decile capital commitments; in columns (4)-(7), a firm is defined as a high reputation firm in a given firm-quarter if its capital commitments is above \$ 20 billion. Finally it includes an interaction between Reputation and $\Delta NAVCF$. All specifications control for PE firm country (acquirer), portfolio firm country (target), portfolio firm's primary industry and Year fixed effects. Standard errors are clustered at the country-pair level.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
$\Delta NAVCF$	-0.00290 (0.0266)	0.0599* (0.0361)	-0.0597 (0.0361)	-0.0140 (0.0161)	-0.00962 (0.0360)	0.0279 (0.0244)	-0.0349 (0.0421)
Reputation* $\Delta NAVCF$	0.0529** (0.0259)	0.102** (0.0419)	0.102** (0.0418)	0.0338* (0.0177)	0.0605* (0.0337)	0.0107 (0.0229)	0.0778* (0.0416)
Reputation	-1.407*** (0.416)	-1.707*** (0.398)	-1.704*** (0.399)	-1.708*** (0.296)	-1.434*** (0.511)	-1.389*** (0.332)	-1.897*** (0.536)
Target_Rev	0.169*** (0.0459)	0.0953* (0.0522)	0.0951* (0.0543)		0.150*** (0.0469)		
Deal_Size		-0.00000272 (0.0000153)				0.00000845 (0.0000183)	
Entry_Multiple			0.000780 (0.000619)				-0.00248* (0.00134)
Pseudo R-squared	0.538	0.501	0.501	0.553	0.542	0.561	0.510
N	2447	1473	1473	10597	2447	3784	1473
Country FE	Y	Y	Y	Y	Y	Y	Y
Industry FE	Y	Y	Y	Y	Y	Y	Y
Year-Quarter FE	Y	Y	Y	Y	Y	Y	Y

Standard errors in parentheses

* $p < .10$, ** $p < .05$, *** $p < .01$